EXP NO: 7

SLIDING WINDOW PROTOCOL

AIM:-

Write a program to implement flow control data link layer using sliding window Protocol. Simulate the flow of frames from one node to another.

SENDER PROGRAM

```
import time
w= int (input ("window size = "))
msg= input ("Message: ")
frames= [(i,c) for i,c in enumerate(msg)]
P=0
while P< len (frames):
  end = min (i+w, len (frames))
  with open ("sender-Buffer.txt","w") as f:
  for n,c in frames [i:end]:
     print ("Frame", n ":", c)
     f.write (f "{n},{c}\n")
  time.sleep(2)
  with open ("Reciever_Buffer.txt") as f:
    ack = f.read().strip()
  if ack.startwith ("ACK:"):
     a = int (ack [4:])
     if a==end
       P= = end
  else:
     print ("wrong ACK-Resending")
  elif ack.startwith ("NACK:"):
```

```
    i=int(ack[5:7])
    print("NACK: Resending form ",i)
print("Transmission complete")
```

RECIEVER PROGRAM:-

```
import time
import os
expected=0
while True:
  if not os.path.exists("sender_Buffer.txt"):
    time.sleep(1)
    continue.
  with open("Send_Buffer.txt") as f:
    frames=f.readlines()
  if not frames:
    time.sleep(1)
    continue.
  error=false
for line in frames:
  n,data=line.strip().split(",")
  n=int(n)
  print("Recieved:",n,data)
  if n!=expected
    with open("Reciever_Buffer.txt","w") as f:
      f.write("NACK:"+str(expected))
    error=True
    break
    expected+=1
```

```
if not error:
    with open("Reciever_Buffer.txt", "w") as f:
        f.write("ACK:" + str(expected))
open("Sender_Buffer.txt", "w").close()
time.sleep(1)
```

SENDER CODE INPUT:

window size: 3

Message: HELLO

SENDER CODE OUTPUT:

FRAME 0 : H

Frame 1 : E

Frame 2 : L

Frame 3 : L

Frame 4 : O

RECIEVER CODE INPUT

sender_Buffer.txt

0, H
1, E
2, L
3, L
4, O

RECIEVER CODE OUTPUT

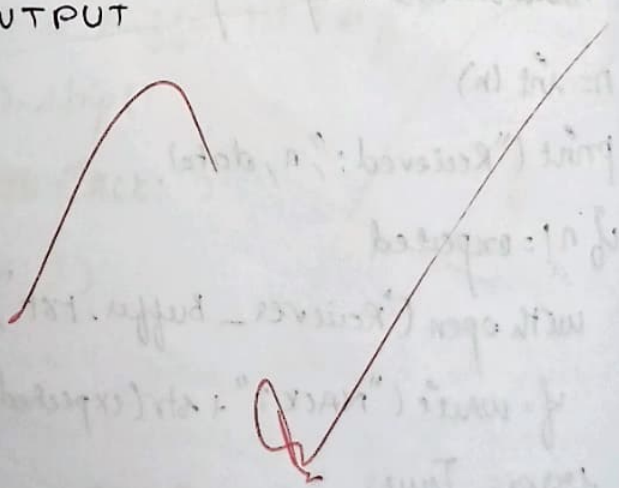
Recieved 0 : H

Recieved 1 : E

Recieved 2 : L

Recieved 3 : L

Recieved 4 : O



RESULT:-

The sliding window Protocol was successfully implemented to control frame flow and handle acknowledgments & retransmissions.